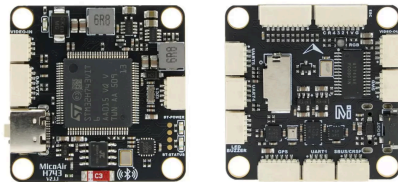


MicoAir743v2

STM32H743VIT6, BMI088+BMI270, SPL06, QMC5883L, OSD, 8xUART, 1xI2C, 1xPWM, 5V 3A BEC, 12V 3A BEC



MicoAir743v2 Flight Controller

High performance H743 flight controller, with dual IMU and bluetooth telemetry built-in, supports Ardupilot/PX4/INAV/Betaflight

Gallery

Specifications

Specifications

- MCU: STM32H743VIT6, 480MHz, 2MB Flash
- IMU: BMI088+BMI270
- Baro: SPL06
- Compass: QMC5883L
- OSD: AT7456E
- MicroSD Card Slot
- 8x UART

- 11x PWM
- 1x I2C
- 1x SWD
- 2x ADC (VBAT, Current)
- USB Type-C
- DJI O3/O4 VTX connector
- BEC 5V 3A output (for controller, receiver, GPS, optical flow or other devices)
- BEC 12V 3A output (for video transmitter, camera)
- Bluetooth Telemetry (connected to UART8 internally, baudrate 115200)
- VBAT Input Range: 2-6S (6-27V)

UART Mapping (Ardupilot)

- SERIAL0 -> USB
- SERIAL1 -> UART1 (MAVLink2, DMA-enabled)
- SERIAL2 -> UART2 (DisplayPort, DMA-enabled)
- SERIAL3 -> UART3 (GPS, DMA-enabled)
- SERIAL4 -> UART4 (MAVLink2, DMA-enabled)
- SERIAL5 -> UART5 (DMA-enabled)
- SERIAL6 -> UART6 (RCIN, DMA-enabled)
- SERIAL7-> UART7 (ESC Telemetry, DMA-enabled)
- SERIAL8 -> UART8 (DMA-enabled, connected to on board BlueTooth Telemetry)

UART Mapping (PX4)

- ttyACM0->USB
- ttyS0->TELEM1 -> UART1
- ttyS1->GPS2 -> UART2
- ttyS2->GPS1 -> UART3
- ttyS3->TELEM2 -> UART4
- ttyS4->TELEM3-> UART5
- ttyS5->RC -> UART6
- ttyS6->URT6 -> UART7
- ttyS7->TELEM4/SERIAL4 -> UART8

RC Input (Ardupilot)

The UART6 is compatible with all ArduPilot supported receiver protocols.

- PPM is not supported.
- SBUS/DSM/SRXL connects to the RX6 pin.
- FPort requires connection to TX6 . See FPort Receivers.
- CRSF also requires a TX6 connection, in addition to RX6, and automatically provides telemetry.

- SRXL2 requires a connection to TX6 and automatically provides telemetry. Set SERIAL6_OPTIONS to “4”.

Any UART can also be used for RC system connections in ArduPilot and is compatible with all protocols except PPM. See Radio Control Systems for details.

OSD Support (Ardupilot)

The MicoAir743v2 supports onboard OSD using OSD_TYPE 1 (MAX7456 driver).

Simultaneously, DisplayPort OSD is available on the HD VTX connector, See below.

VTX Support

The SH1.0-6P connector supports a DJI Air Unit / HD VTX connection. Protocol defaults to DisplayPort. Pin 1 of the connector is 12v so be careful not to connect this to a peripheral requiring 5v.

PWM Output

The MicoAir743v2 supports up to 11 PWM outputs.

Channels 1-8 support DShot and bi-directional DShot.

(currently PX4&INAV not support BDSHOT function)

PWM output share grouped and every group must use the same output protocol:

1,2,3,4 are group 1

5,6 are group 2

7,8,11 are group 3

9,10 are group 4

Note: PWM11 is the “LED” pin. If this is configured for serial LED use then PWM 7 and 8 can only be used as serial LED also.

Battery Monitoring

The board has a internal voltage sensor and connections on the ESC connector for an external current sensor input.

The voltage sensor can handle up to 6S LiPo batteries.

The default battery parameters(for Ardupilot) are:

- BATT_VOLT_PIN 10
- BATT_CURR_PIN 11
- BATT_VOLT_MULT 21.2
- BATT_CURR_SCALE 40.2

Compass

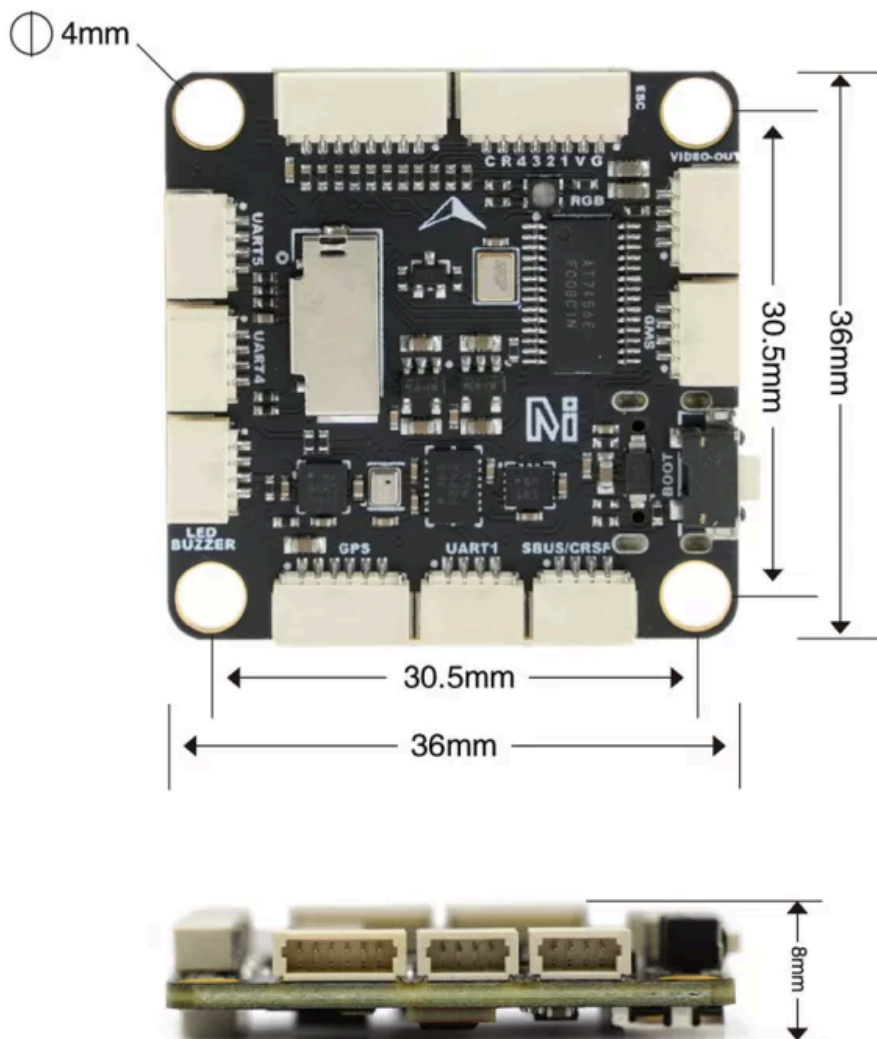
The MicoAir743v2 has a built-in compass sensor (QMC5883L), and you can also attach an external compass using I2C on the SDA and SCL connector.

Physical

- Mounting: 30.5 x 30.5mm, Φ 4mm
- Dimensions: 36 x 36 x 8 mm
- Weight: 10g

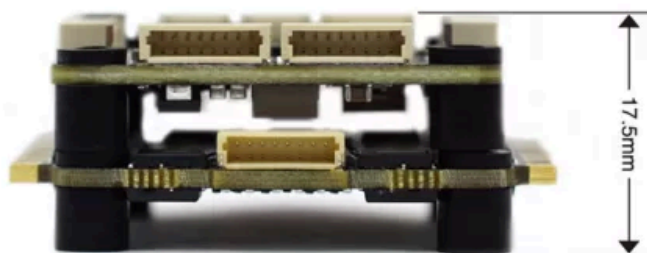
Size

MicoAir743 V2



◆ Weight: 10g

MicoAir743 V2 Flight Stack



◆ Weight: 24g

Related-links

- [Mission Planner](#)
- [QGround Control](#)
- [ArduPilot Documentation](#)
- [PX4 Documentation](#)
- [Betaflight Configurator](#)
- [INAV Configurator](#)

Ports

Diagram



BlueTooth

Pinout

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